

Brief Introduction to NoSQL

MongoDB & Cosmos DB Awareness for JSON Data

1. What is NoSQL?

NoSQL stands for 'Not Only SQL'. These databases are designed to store and manage large volumes of semi-structured, unstructured, and JSON-based data.

2. Why NoSQL Became Popular?

Modern applications such as Netflix, Amazon, Uber, and Instagram generate huge amounts of real-time and flexible data. Traditional relational databases become difficult to scale in such scenarios, leading to the popularity of NoSQL databases.

3. Features of NoSQL Databases

- Flexible Schema
- High Scalability
- JSON Document Support
- Faster Development
- Real-Time Processing

4. Types of NoSQL Databases

- Document Database – MongoDB, Cosmos DB
- Key-Value Store – Redis
- Column Family – Cassandra
- Graph Database – Neo4j

5. What is a Document Database?

A document database stores data in JSON-like documents instead of rows and columns.

Example JSON Document

```
{  
  "customer_id": 101,  
  "name": "Anand",  
  "city": "Surat",  
  "skills": ["SQL", "Python", "Snowflake"]  
}
```

6. Introduction to MongoDB

MongoDB is one of the most popular NoSQL document databases. It stores data in BSON internally and JSON-like documents externally.

MongoDB Structure

SQL Database → MongoDB

Table → Collection

Row → Document

Column → Field

7. Advantages of MongoDB

- Easy to scale
- Flexible structure
- Cloud-friendly
- Handles JSON naturally
- Suitable for real-time systems

8. What is Azure Cosmos DB?

Azure Cosmos DB is Microsoft's globally distributed cloud-based NoSQL database service designed for massive scalability and low latency.

9. Features of Cosmos DB

- Global Distribution
- Multi-Model Support
- Fully Managed Service
- High Performance

10. Cosmos DB APIs

- SQL API
- MongoDB API
- Cassandra API
- Gremlin API

11. SQL vs NoSQL

SQL:

- Structured Tables
- Fixed Schema
- Strong Relationships

NoSQL:

- JSON/Documents
- Flexible Schema
- Horizontally Scalable

12. Real-World Use Cases

- E-commerce – Product Catalogs
- Social Media – User Activity

- IoT Systems – Sensor Data
- OTT Platforms – Recommendations

13. Final Takeaway

NoSQL databases are ideal for modern cloud-native applications, real-time systems, and JSON-based flexible data storage. Popular examples include MongoDB and Cosmos DB.